Manav Mepani

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EDUCATION

Worcester Polytechnic Institute (WPI), USA **Aug 2024 – May 2026**
Master of Science in Robotics, **CGPA – 4.0/4.0**

Sardar Patel College of Engineering (SPCE), India **Aug 2018 – Jun 2022**
Bachelor of Technology in Mechanical Engineering, **CGPA – 9.04/10**

EXPERIENCE

EssilorLuxottica – GENTEX (USA) | *Automation Support Intern* **May 2025 - Present**

- Improved automated label-inspection accuracy and reduced customer mispack complaints by developing and optimizing computer-vision checks using the Keyence Vision System.
- Migrated Mitsubishi robot code to Epson, refactored for clarity, redesigned the homing sequence, and integrated robot control with Studio 5000 PLCs for synchronized line operation.
- Produced clear technical documentation and worked extensively with industrial automation systems, including Magnemotion carriers, 6-axis robots, SCARA, and gantry systems.

Key skills: Robot Programming (Epson RC+, Mitsubishi), Vision Systems (Keyence), Documentation & Technical Writing

The Innovation Story (India) | *Robotics Engineer* **Aug 2023 – Apr 2024**

- Designed and manufactured complex systems(🔗) for game challenges, achieving **~90% reliability** within three months using innovative designs and in-house manufacturing.
- Worked with teams of 20 students of various grades while teaching them physics, CAD, and manufacturing for the competition in which they stood **4th all over India**.

Key Skills: Python Programming, Java Programming, CAD Designing, Design for manufacturing, Teamwork, Mechatronics

Technology Innovation Hub (India) | *Project Research Assistant* **Jun 2022 – Aug 2023**

- Designed and manufactured an autonomous onion harvester(🔗) for small farms, featuring an **Energy-Aware Path Planning algorithm** and a **return-to-base function** triggered by low battery levels to ensure operational efficiency.
- **Improved the maneuverability of a bomb disposal rover** by using a design that ensures at least one side's wheels maintain ground contact for enhanced stability.
- Optimized **rocker-bogie mechanism** design for a rover using **Simulated Annealing optimization** algorithm in Octave to enhance step-climbing ability by analyzing key design parameter relationships.

Key Skills: Python, Octave, optimization algorithm, CAD Designing, Design for manufacturing, FEA, Teamwork, Problem solving

SOFTWARE AND TECHNICAL COMPETENCIES

Softwares: CATIA V5/SOLIDWORKS, AutoCAD, ANSYS Workbench, ADAMS, FluidSIM, MS OFFICE SUIT, Studio 5000 (Allen-Bradley)

Programming Skills: C++, Arduino, Python, Octave, Java, Webots, Robot Operating System (ROS2), MATLAB, Epson RC+ (SPEL+), RT ToolBox3.

Certifications(🔗)**:** Neural Networks and Deep learning, Python Programming for data structure and accessing web data, Digital Electronic circuits, ANSYS Workbench, PLC

ACADEMIC PROJECTS

Chance Constrained Rapid-Exploring Random Tree(🔗) **March 2025 – May 2025**

- Implemented a Chance-Constrained RRT motion planner accounting for Gaussian uncertainty, probabilistic collision checks, and dynamic obstacles, using Docker and GitHub for reproducible CI/CD-style development and version control.
- Designed and executed simulation experiments validating uncertainty-aware, collision-safe trajectories, achieving high planning reliability across varied obstacle configurations.

Robotic Arm Control Projects using ROS 2(🔗) **Oct 2024 – Dec 2024**

- Designed Python-based ROS 2 nodes to calculate velocity-level kinematics, implement incremental position references, and execute linear path motions for a robotic arm.
- Utilized Jacobian matrices and pseudoinverse methods to calculate joint and end-effector velocities, enabling precise motion control in Cartesian space.

Undergrad Thesis: Design and Development of an Autonomous Cooking System(🔗) **Aug 2021 – May 2022**

- Designed, fabricated and analysed the design for a 6 DOF **PUMA-configured** robotic arm for autonomous cooking.
- Simulated **Inverse Kinematics algorithm** which used **kinematic decoupling method** in **Webots** and controlled the physical model using an Arduino microcontroller.
- Achieved a **0.7 Kg (1.55 lbs) payload capacity** with moderately high repeatability of **±1° error** in motion control using PID.